CSMMagEco2026 — New US Magnet Companies Identified

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** `CSMMagEco2026 — New US Magnet Companies Identified V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
DFARS deadline status	2027 deadline referenced; implementation guidance pending	Implementation guidance released Q1 2026; supply chain disclosure required by Sep 2026	6 months closer to enforcement; actionable compliance steps now published	DoD DFARS Q1 2026 guidance
Niron $Fe_{16}N_2$ performance	BHmax 24 MGOe demonstrated	BHmax 27 MGOe on production-scale samples (2025 update)	Direct improvement changes feasibility assessment for non-REE motor applications	Niron Magnetics 2025 data release
Urban Mining Co capacity	500 kg/month processing	12,000 kg/month facility online Q2 2026, Pflugerville TX	24× scale-up changes market significance; now DFARS-certified source	Urban Mining Company Q2 2026 announcement

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MP Materials magnet production	Mining only; manufacturing partnership announced	Fort Worth TX sintered NdFeB facility online Q4 2025 — first US NdFeB in 30+ years	V1.0 manufacturing gap now partially resolved — significant change to supply assessment	MP Materials Q4 2025 announcement
Dy/Tb price	Reference price 2024	Dy $4,100/kg (May 2026); Tb $6,200/kg — China ECA restriction impact	Price data directly affects project cost estimates and substitution economics	Reuters metals data May 2026
MXene magnet nanocomposites	Not covered in V1.0	MXene $Ti_3C_2T_x$ admixed NdFeB: +12% coercivity, +25°C operating temp (Drexel 2025)	Emerging technology relevant to Aegis high-temperature motor magnet requirements	Drexel University 2025 preprint
DoD demand projection	General demand cited	CBO 2026 study: DoD magnet requirement +340% by 2030	Quantitative demand data enables production scaling analysis V1.0 lacked	Congressional Budget Office 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
DFARS deadline status	6 months closer to enforcement; actionable compliance steps now published
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MP Materials magnet production	V1.0 manufacturing gap now partially resolved — significant change to supply assessment

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Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMMagEco2026 — New US Magnet Companies Identified V1.0 archived; V2.0 is the active specification as of June 2026.